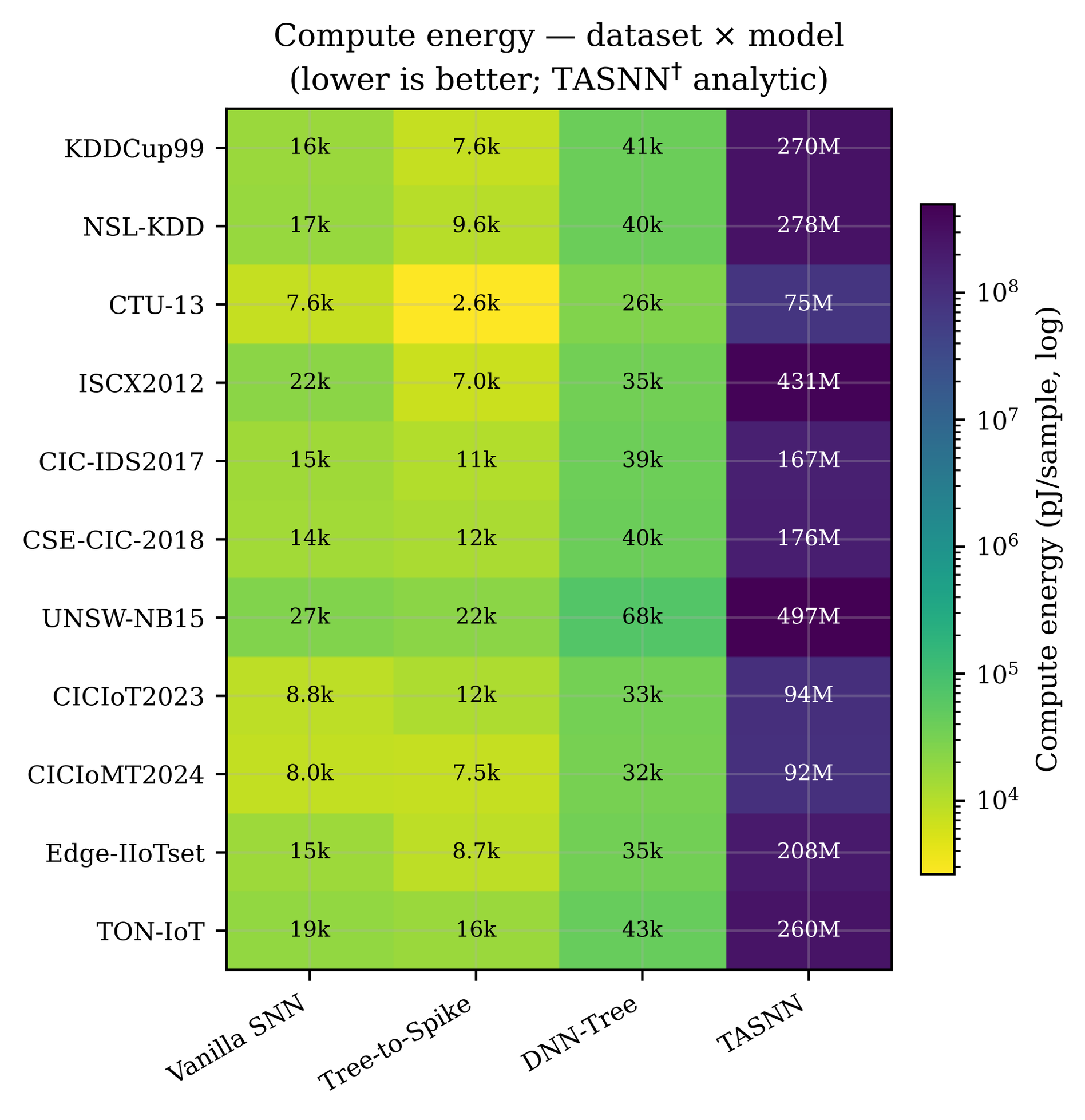
Compute energy — dataset × model
(lower is better; TASNN† analytic)
KDDCup99
NSL-KDD
CTU-13
ISCX2012
CIC-IDS2017
CSE-CIC-2018
UNSW-NB15
CICIoT2023
CICIoMT2024
Edge-IIoTset
TON-IoT
16k
7.6k
41k
270M
17k
9.6k
40k
278M
7.6k
2.6k
26k
75M
22k
7.0k
35k
431M
15k
11k
39k
167M
14k
12k
40k
176M
27k
22k
68k
497M
8.8k
12k
33k
94M
8.0k
7.5k
32k
92M
15k
8.7k
35k
208M
19k
16k
43k
260M
Vanilla SNN
Tree-to-Spike
DNN-Tree
TASNN
Compute energy (pJ/sample, log)
10^8
10^7
10^6
10^5
10^4